PyGame Flappy Bird Tutorial - Part 5

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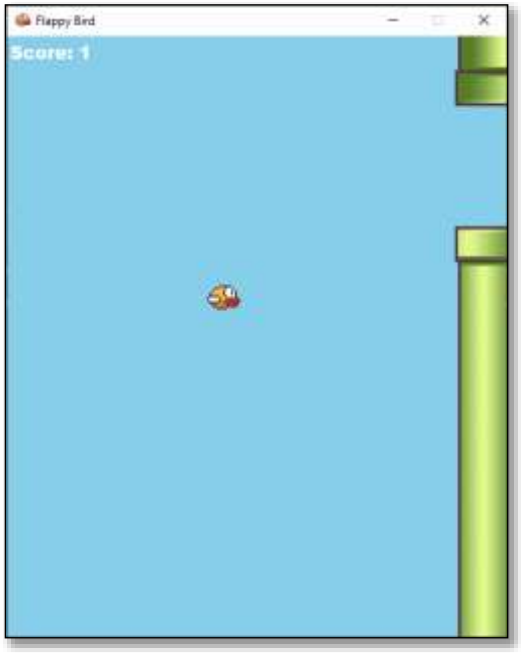
Time required: 30 minutes

Preview of the Game

Here's a sneak peak of the game that we are going to work on.

[Flappy Bird Demo Video]



Update Background

Save **flappy_bird_4.py** as **flappy_bird_5.py**

Time to get our background moving. This will help with the illusion of movement.

Add self.background_speed to the class.

```
        # Set the gravity to 3
        # This is how fast the bird falls
        # The higher the number, the faster the bird falls
        # The lower the number, the slower the bird falls
        self.gravity = 3
        self.pipes_speed = 4  # Pipes move faster than background
        self.background_speed = 2  # Background moves slower than pipes

        self.load_background()
        self.init_bird()
        self.init_pipes()
```

Add the **update_background()** method.

```
    # --------------------- UPDATE BACKGROUND ---------------------------- #
    def update_background(self):
        """Move background image seamlessly"""
        # Move the background image to the left by self.background_speed
        self.background_rect.left -= self.background_speed

        # If the first background image is off the left side of the screen
        if self.background_rect.right <= 0:
            # Reset it to the right side of the screen
            self.background_rect.left = 0
```

Reset the Pipes

We only get one set of pipes. Let's reset the pipes to have more pipes traveling across.

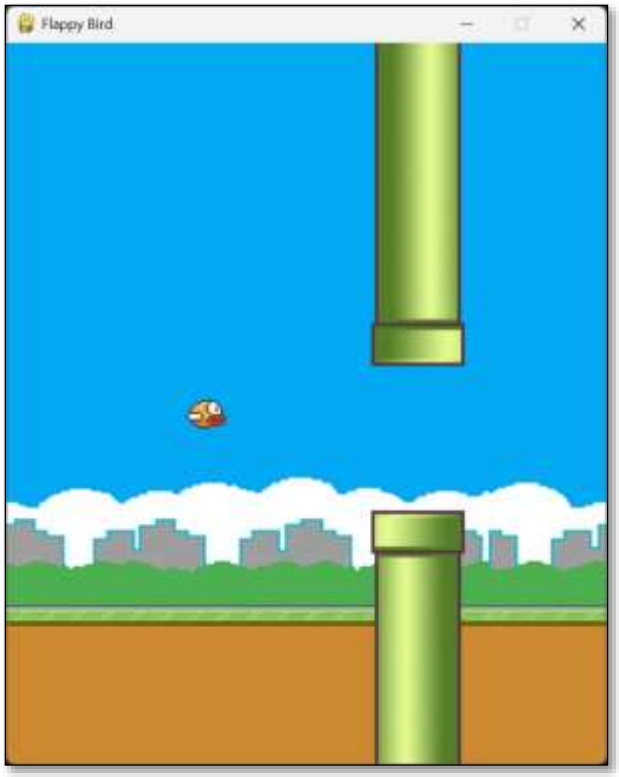
Let's add a **reset_pipes** method.

```python
    # -------------------- RESET PIPES -----------------------------------#
    def reset_pipes(self):
        """Reset pipes every time they leave the screen"""
        # Pick a random height for the bottom of the top pipe
        self.pipe_upper_rect.bottom = randint(
            50,                 # Set minimum random number to 50
            config.HEIGHT // 2  # Set maximum to half the surface height
        )

        # Set lower pipe top to upper pipe bottom plus pipe gap
        self.pipe_lower_rect.top = self.pipe_upper_rect.bottom \
            + self.pipe_gap_size

        # Set initial X off screen to right
        self.pipe_upper_rect.left = config.WIDTH
        self.pipe_lower_rect.left = config.WIDTH
```

Update Game Loop

```python
    # ---------------------------- GAME LOOP ------------------------------ #
    def game_loop(self):
        """Infinite game loop"""
        while True:
            self.check_events()
            self.update_bird()
            self.update_pipes()
            self.update_background()

            # If the pipes are off the screen, reset them
            if self.pipe_upper_rect.right < 0:
                self.reset_pipes()
            # ------------------- DRAW SURFACE ---------------------------- #
            # Filling the surface with the background image
            # clears the previous frame
            # Draw the background
            self.surface.blit(self.background, self.background_rect)
            self.surface.blit(self.background, (self.background_rect.right, 0))

            # Draw bird to the surface
            self.surface.blit(self.bird, self.bird_rect)

            # Draw pipes to the surface
            self.surface.blit(
                self.pipe_lower,  # Source image
                self.pipe_lower_rect,  # Destination location of image
            )
            self.surface.blit(
                self.pipe_upper,  # Source image
                self.pipe_upper_rect,  # Destination location of image
            )

            # ----------------- UPDATE DISPLAY ---------------------------- #
            # From surface, update Pygame display to reflect any changes
            pygame.display.update()
            # Cap game speed at 60 frames per second
            self.clock.tick(60)
```

Example run:



You can fly your bird up and down and through the pipes. The pipes keep coming!

There are a few issues. The bird can fall off the screen or fly up to the sun. There aren't any collisions or score keeping.

Coming right up!

Assignment Submission

1. Attach a screenshot showing the operation of the program.
2. Zip up the program files folder and submit in Blackboard.